Data Selection and Training Budget for Reasoning Supervised Fine-Tuning of Qwen2.5-3B-Instruct

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Abstract

Supervised fine-tuning (SFT) on long chain-of-thought (CoT) demonstrations is a common way to improve the reasoning ability of small language models, but which demonstrations to train on remains an open design choice. We study this choice for Qwen2.5-3B-Instruct under a fixed budget of 15K examples drawn from a 100K subset of AceReason-1.1-SFT. We design a label-free selection pipeline that combines domain filtering, exact deduplication, length filtering to the training context, and stratified sampling over data source and solution length. On eight math and science benchmarks, our final model, trained for five epochs on the selected data, achieves a macro pass@8 of 59.0% on the six leaderboard benchmarks, compared with 53.9% for the base model and 54.9% for a random-selection baseline, as well as the highest macro pass@1 (26.9%). Our analysis shows that the gains stem primarily from fewer generations being truncated at the 32,768-token limit.

1 Introduction

Distilling long CoT traces from strong reasoning models into smaller models through SFT has become a standard recipe for improving mathematical reasoning [Guha et al., 2026, Liu et al., 2025, Muennighoff et al., 2025]. Small models, however, do not benefit uniformly from such data: they often fail to absorb long reasoning traces [Li et al., 2025], and long-reasoning models are prone to overthinking, i.e., generating excessively long reasoning [Chen et al., 2024]. The choice of training examples therefore matters when the data budget is limited.

In this project [reds-lab, 2026], we fine-tune Qwen2.5-3B-Instruct [Yang et al., 2024] on 15K examples selected from a 100K subset of AceReason-1.1-SFT [Liu et al., 2025], whose responses are long CoT traces generated by DeepSeek-R1, and ask whether a principled selection of the 15K examples outperforms random selection. Our contributions are as follows:

- We design a label-free data selection pipeline that requires no model inference and yields complete, deduplicated, and length-balanced math demonstrations (Section 2).
- Our final model, trained for five epochs on the selected data, improves macro pass@8 by 5.1 points over the base model and macro pass@1 by 2.3 points over the random baseline; both 95% confidence intervals exclude zero (Section 4).
- We identify truncation at the generation limit as the dominant failure mode of the fine-tuned models and separate the contributions of data selection and training budget (Section 4).

2 Data selection strategy

Random baseline. The training pool contains 100,000 examples, of which 67,432 are math and 32,568 are code. The random baseline samples 15,000 examples uniformly with a fixed seed and

Table 1: Training hyperparameters of the final model and the random baseline.

	Final model	**Random baseline**
Training data	Selected 15K (107.6M tokens/epoch)	Random 15K (113.4M tokens/epoch)
Epochs (steps)	5 (9,375)	3 (5,625)
Learning rate	5×10^{-6}, cosine, no warmup; epochs 4–5 decay from 1.7×10^{-6} to 0	5×10^{-6}, cosine, no warmup
Tokens seen	538.2M	338.8M
Hardware (time)	4×L40S (8.6 h) + 4×A100 (4.4 h)	4×L40S (9.2 h)
Base model	Qwen2.5-3B-Instruct, full-parameter SFT, loss on response tokens	
Optimizer	AdamW, $\beta = (0.9, 0.999)$, weight decay 0, gradient clipping 1.0	
Global batch size	8 (4 GPUs × micro-batch 1 × gradient accumulation 2)	
Max. sequence length	16,384 tokens, Qwen chat template	
Precision, kernels	bf16, DeepSpeed ZeRO-3, FlashAttention-2, Liger kernel	

no filtering. Of these, 556 exceed the 16,384-token training context and are truncated, so their final answers are removed from the training target.

Selected set. Our pipeline applies four steps in order. First, *domain filtering* keeps only math examples, since all target benchmarks are math or science and code traces are off-target. Second, *exact deduplication* removes repeated prompts (7,730 among the math examples) and keeps the first occurrence, which increases prompt diversity [Liu et al., 2025]. Third, *length filtering* keeps only examples whose full conversation fits the 16,384-token context under the Qwen chat template (removing 2,292), so that every training target ends with its final answer and end-of-turn token. Fourth, *stratified sampling* with proportional allocation draws 15,000 of the 57,410 remaining examples over strata defined by data source (NuminaMath-CoT and OpenMathReasoning) and total-length band (up to 4K, 4–8K, and 8–16K tokens), which preserves the source mix and the range of difficulty implied by solution length. The resulting set contains 107.6M training tokens per epoch (113.4M for the random set), has a median length of 6.5K tokens, and shares 2,252 examples with the random set.

Design rationale. The pipeline follows evidence that a small, curated fine-tuning set can match or exceed a larger unfiltered one, both for alignment [Zhou et al., 2023] and for reasoning [Muennighoff et al., 2025, Ye et al., 2025], as well as the question filtering and source mixing studied in data recipes for reasoning models [Guha et al., 2026]. It uses no model scoring, embeddings, or answer keys. A data-contamination check found six pool prompts that match a benchmark question verbatim; one of them falls into each 15K set and is kept and disclosed rather than removed from only one arm.

3 Experimental setup

Training. Both models are trained with full-parameter SFT in LLaMA-Factory [Zheng et al., 2024], using DeepSpeed ZeRO-3 [Rajbhandari et al., 2020], FlashAttention-2 [Dao, 2024], and the Liger fused linear cross-entropy kernel [Hsu et al., 2024]; the loss is computed on response tokens only. Table 1 lists the hyperparameters. The final model is trained for three epochs and then resumed from the epoch-3 weights, optimizer, and scheduler states for two additional epochs on a five-epoch cosine schedule.

Benchmarks and metrics. We evaluate on AIME24, MATH-500, CN-Math 2024, Kaoyan, AMC, Minerva, OlympiadBench, and GPQA using the course evaluation harness, which generates with vLLM [Kwon et al., 2023] and grades the last boxed answer by symbolic equivalence. For each question, we draw $n = 8$ samples with temperature 0.6, top-p 0.95, and a maximum of 32,768 generated tokens. We report pass@8, which counts a question as solved if any of the eight samples is correct (the pass@k metric of Chen et al. [2021] with $k = n$), and pass@1, the mean per-sample accuracy. The macro score averages, with equal weight, the six leaderboard benchmarks available to us (CN-Math 2024, Kaoyan, AMC, Minerva, OlympiadBench, and GPQA); AIME24 and MATH-500 are reported but excluded from the leaderboard average. We report differences between models in percentage points with 95% confidence intervals (CIs) from a paired bootstrap over questions

Table 2: Results on the eight benchmarks (pass@8 and pass@1, %). N: number of questions; *: benchmark included in the macro average.

	Base		Random 15K		Final model	
Benchmark (N)	pass@8	pass@1	pass@8	pass@1	pass@8	pass@1
AIME24 (30)	16.7	7.1	33.3	7.1	23.3	7.5
MATH-500 (500)	84.0	65.3	88.4	64.7	87.8	66.0
CN-Math 2024* (30)	36.7	10.4	33.3	7.9	33.3	10.4
Kaoyan* (199)	54.8	21.9	65.3	29.2	67.8	29.8
AMC* (40)	65.0	39.1	57.5	34.4	77.5	38.1
Minerva* (272)	45.2	28.7	47.1	25.5	45.6	27.3
OlympiadBench* (675)	49.8	27.7	52.6	27.8	53.8	30.1
GPQA* (198)	71.7	29.4	73.7	23.0	75.8	25.8
Macro (*)	53.9	26.2	54.9	24.6	**59.0**	**26.9**

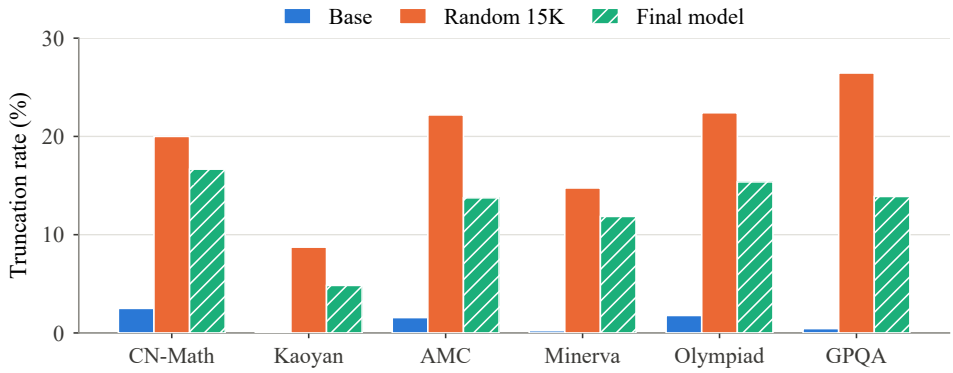


Figure 1: Truncation rate, i.e., the share of samples that reach the 32,768-token generation limit, on the six leaderboard benchmarks.

($B = 2000$), in which the eight samples of each question are kept together [Efron and Tibshirani, 1993, Miller, 2024].

4 Results and analysis

Main results. Table 2 summarizes the results. Fine-tuning on randomly selected data yields only a marginal improvement over the base model (+1.1 points in macro pass@8, 95% CI [−4.2, 6.2]) and even reduces macro pass@1 by 1.6 points. In contrast, our final model improves macro pass@8 by 5.1 points over the base model (95% CI [0.9, 9.5]), the only improvement over the base model that is significant at the 95% level, and outperforms the random baseline by 4.0 points in pass@8 (95% CI [−0.4, 8.6]) and by 2.3 points in pass@1 (95% CI [0.9, 3.8]). The gains are most pronounced on Kaoyan (+13.1 points in pass@8 over the base model) and AMC (+20.0 points over the random baseline). On AIME24, which is excluded from the macro average, the final model underperforms the random baseline by 10.0 points, i.e., three of 30 questions.

Truncation explains most of the gain. We find that truncation at the generation limit is the dominant failure mode of the fine-tuned models. Both fine-tuned models generate CoT traces 8–16× longer than the base model, consistent with the overthinking behavior reported for long-reasoning models [Chen et al., 2024]. As shown in Figure 1, 8.7–26.5% of the samples of the random baseline reach the 32,768-token limit before producing an answer, and such truncated samples are almost never correct (0–4% accuracy). Our final model reduces the truncation rate on every leaderboard benchmark, to 4.8–16.7%, while the accuracy of completed samples remains comparable (e.g., 35.5% vs. 35.2% on OlympiadBench and 30.9% vs. 29.6% on Minerva). Reducing truncation therefore translates directly into higher pass@1 and pass@8. We hypothesize that two factors contribute. First, length filtering ensures that every training target is a complete solution that ends with an answer,

whereas 556 targets in the random set are truncated. Second, longer training further suppresses run-on generation: on a held-out set of 200 pool questions that appear in neither training set, the truncation rate of the selected run decreases from 55% after epoch 1 to 24% after epoch 3 and 16% after epoch 5. This is consistent with Liu et al. [2025], who report that SFT accuracy on this data family keeps improving up to about five epochs.

Data selection or training budget? Because the final model is trained for five epochs and the random baseline for three, the comparison conflates data selection and training budget. Under an equal budget of three epochs, the selected data achieves a macro pass@8 of 56.6% and pass@1 of 24.4%, a gain of 1.6 points in pass@8 over the random baseline (95% CI $[-2.4, 5.6]$) that lies within question-sampling noise. We therefore attribute the improvement of the final model to the combination of the selected data and the longer training, rather than to data selection alone.

Limitations. Each configuration is trained once, so our confidence intervals capture question-sampling variance but not training-seed variance. The five-epoch schedule was chosen after the three-epoch results were known, and the final model is selected on the benchmarks on which it is reported; AIME25, which is part of the course leaderboard but not available to us, is the only unseen test set. All training data come from the provided pool. Adding external data is allowed but would require decontamination against the benchmarks; short, curated reasoning sets [Muennighoff et al., 2025, Ye et al., 2025], which suit small students [Li et al., 2025], and science data for GPQA are natural extensions. Additional selection methods we evaluated, including a learning-progress selector, are documented in the full report in our repository.

5 Conclusion

We studied training-data selection for reasoning SFT of Qwen2.5-3B-Instruct under a 15K-example budget. A label-free selection of complete, deduplicated math demonstrations, combined with five epochs of training, raises macro pass@8 from 53.9% to 59.0% and macro pass@1 from 26.2% to 26.9%, outperforming the random baseline (54.9% and 24.6%). Our analysis suggests that, for a 3B student trained on long CoT data, reducing truncated generations is a key lever for improving reasoning performance. Code, configurations, and evaluation logs are available at `https://github.com/yuan-zhou-git/Course_Project_Trustworthy_ML`, and the final model and datasets at `https://huggingface.co/yuan-zhou-hf`.

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